at.sea.lobster

An Application for at-sea sampling data entry and quality checks

Before you Begin *(any line starting with > is code; only type everything after the >)

1. Download Rstudio
https://rstudio-education.github.io/hopr/starting.html

3. Download Devtools in R Studio
https://cran.r-project.org/web/packages/devtools/readme/README.html

> install.packages ("devtools")

4. Review the ReadMe content for the package
https://github.com/LobsterScience/at.sea.lobster

5. Resources for species codes and data sheets available on google drive.
https://drive.google.com/drive/folders/1X00vXglwr8_Dg6A_TXP3n3FhE2laUK1f

6. Install the at.sea.lobster App and load package

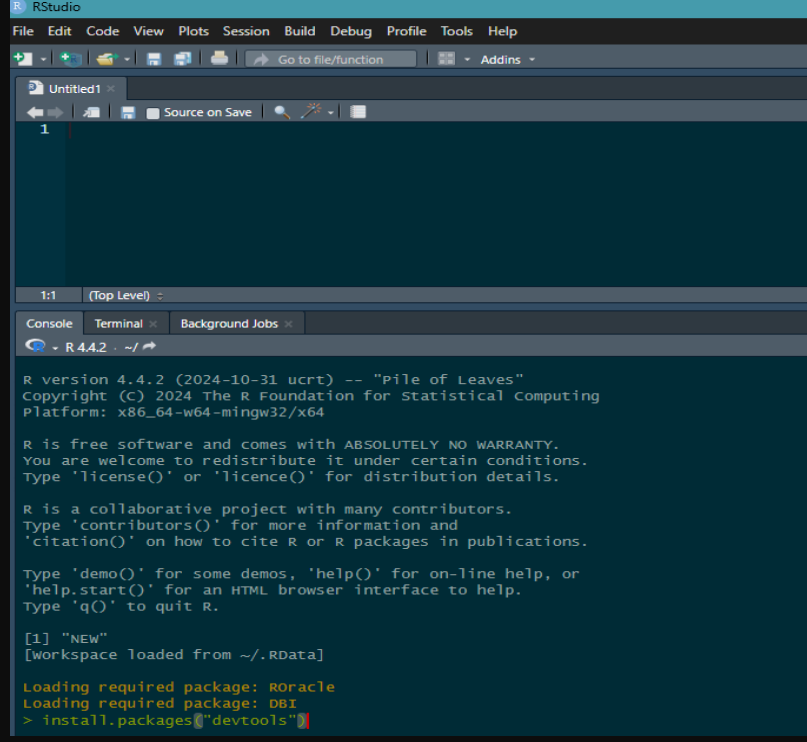
> devtools::install_github("LobsterScience/at.sea.lobster")

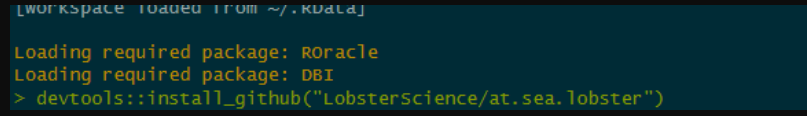
> require(at.sea.lobster)

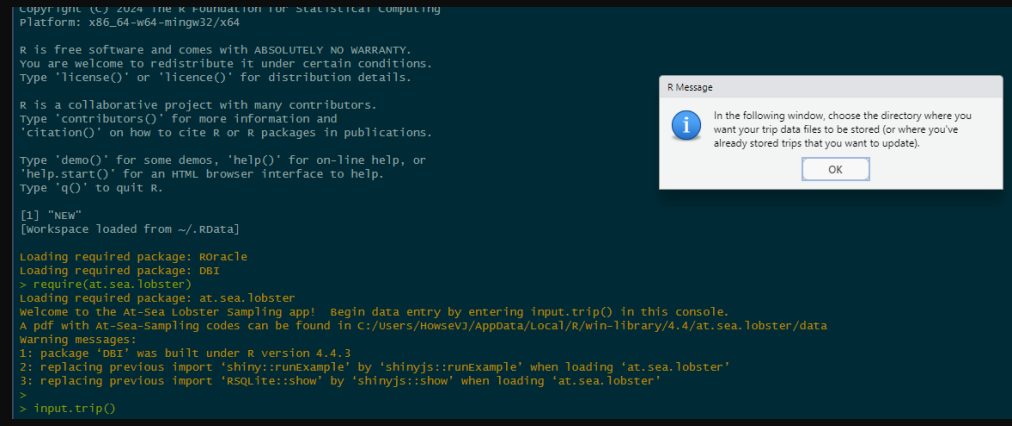
7. Open window to enter data

> input.trip()

(pick the folder where you want to save your datafiles)



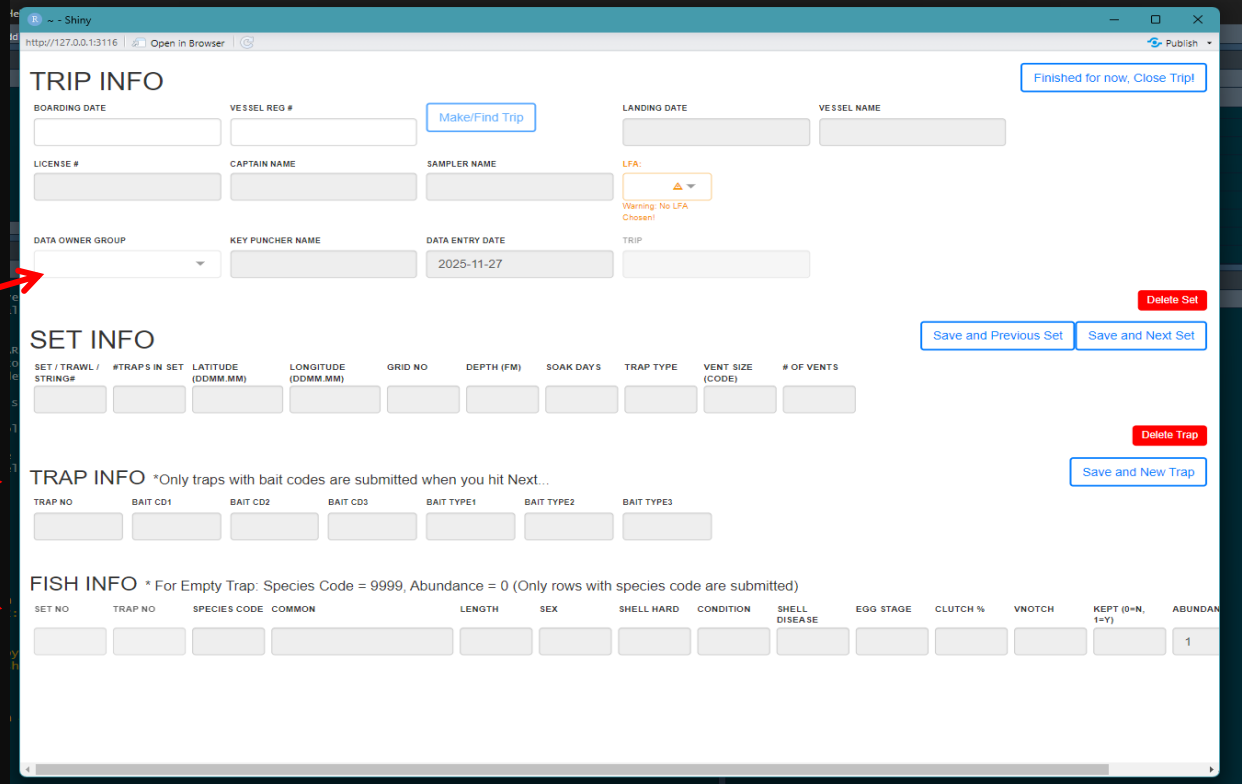


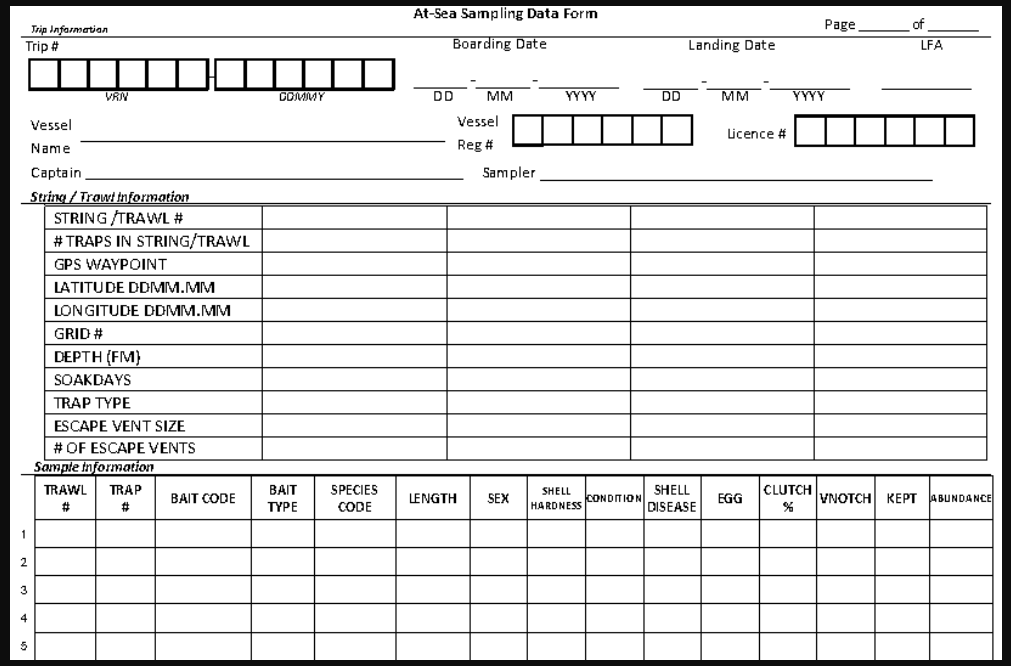


Overview of input.trip() screen:

- The data entry system is set up to mirror the data collection sheets used on the boat.

- Each TRIP will have header information that will not change
- A TRIP has multiple SETS
  - A SET has multiple TRAPS
    - A trap has multiple FISH (or animals)





At-Sea Sampling Data Form

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Trip Information

Trip # ____ - ____ (VRN - DDMMY)

Boarding Date ____-____-____ (DD MM YYYY)

Landing Date ____-____-____ (DD MM YYYY)

LFA _____

Vessel Name _____

Vessel Reg # _____

Licence # _____

Captain _____

Sampler _____

String / Trawl Information

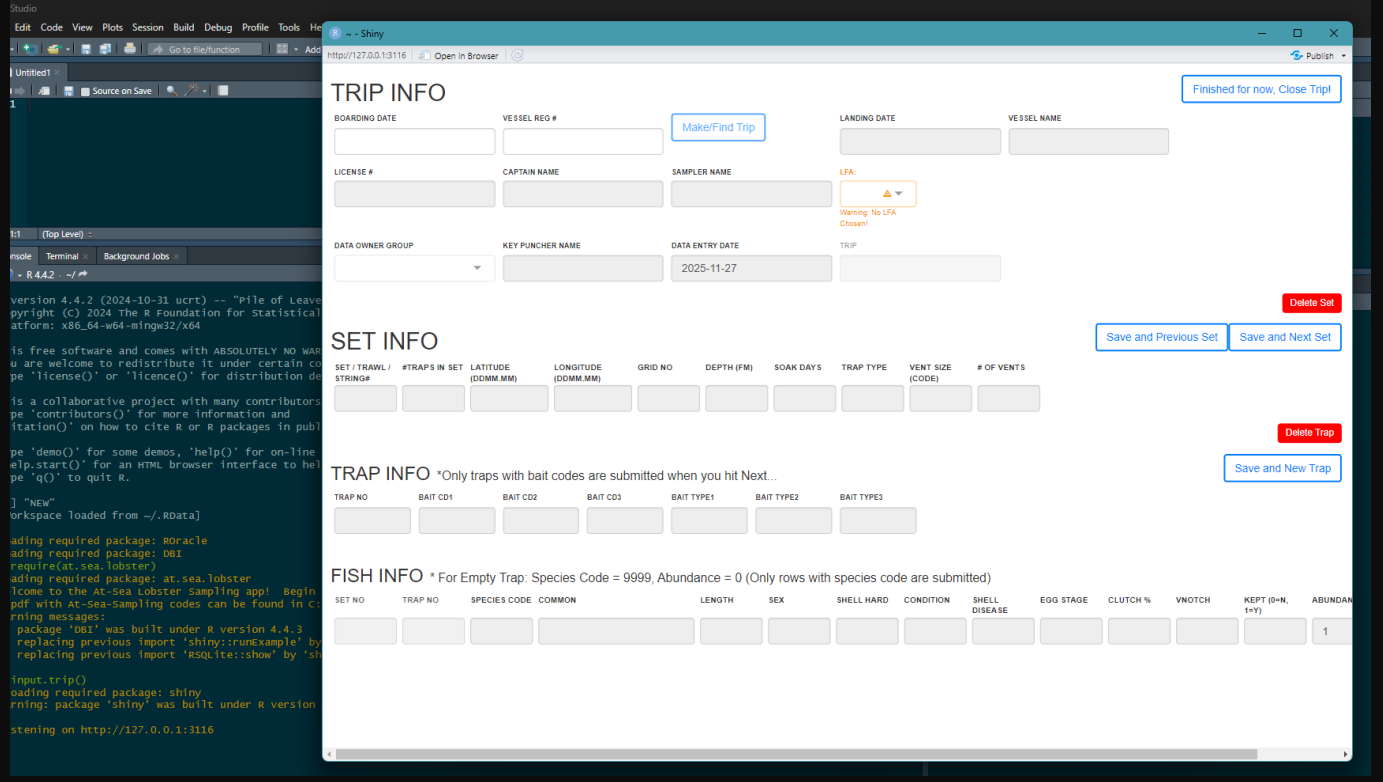
STRING /TRAWL #				
# TRAPS IN STRING/TRAWL				
GPS WAYPOINT				
LATITUDE DDMM.MM				
LONGITUDE DDMM.MM				
GRID #				
DEPTH (FM)				
SOAKDAYS				
TRAP TYPE				
ESCAPE VENT SIZE				
# OF ESCAPE VENTS				

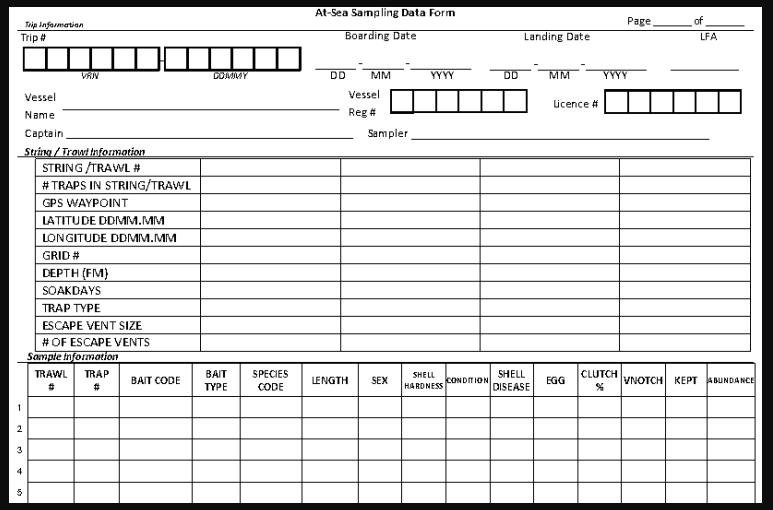
Sample Information

	TRAWL #	TRAP #	BAIT CODE	BAIT TYPE	SPECIES CODE	LENGTH	SEX	SHELL HARDNESS	CONDITION	SHELL DISEASE	EGG	CLUTCH %	VNOTCH	KEPT	ABUNDANCE
1															
2															
3															
4															
5															

Enter Your Data:

1. Once a folder is selected the Data Entry window will open.
   - This folder is where all your entered data will be saved. Please ensure you keep track of the location of this folder on your computer and be consistent in the folder you are using.





At-Sea Sampling Data Form

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Trip Information

Trip # ☐☐☐☐☐-☐☐☐☐☐ (VRN, DDMMY) Boarding Date ___-___-___ (DD MM YYYY) Landing Date ___-___-___ (DD MM YYYY) LFA _____

Vessel Name _____ Vessel Reg # ☐☐☐☐☐☐ Licence # ☐☐☐☐☐☐

Captain _____ Sampler _____

String / Trawl Information

STRING /TRAWL #				
# TRAPS IN STRING/TRAWL				
GPS WAYPOINT				
LATITUDE DDMM.MM				
LONGITUDE DDMM.MM				
GRID #				
DEPTH (FM)				
SOAKDAYS				
TRAP TYPE				
ESCAPE VENT SIZE				
# OF ESCAPE VENTS				

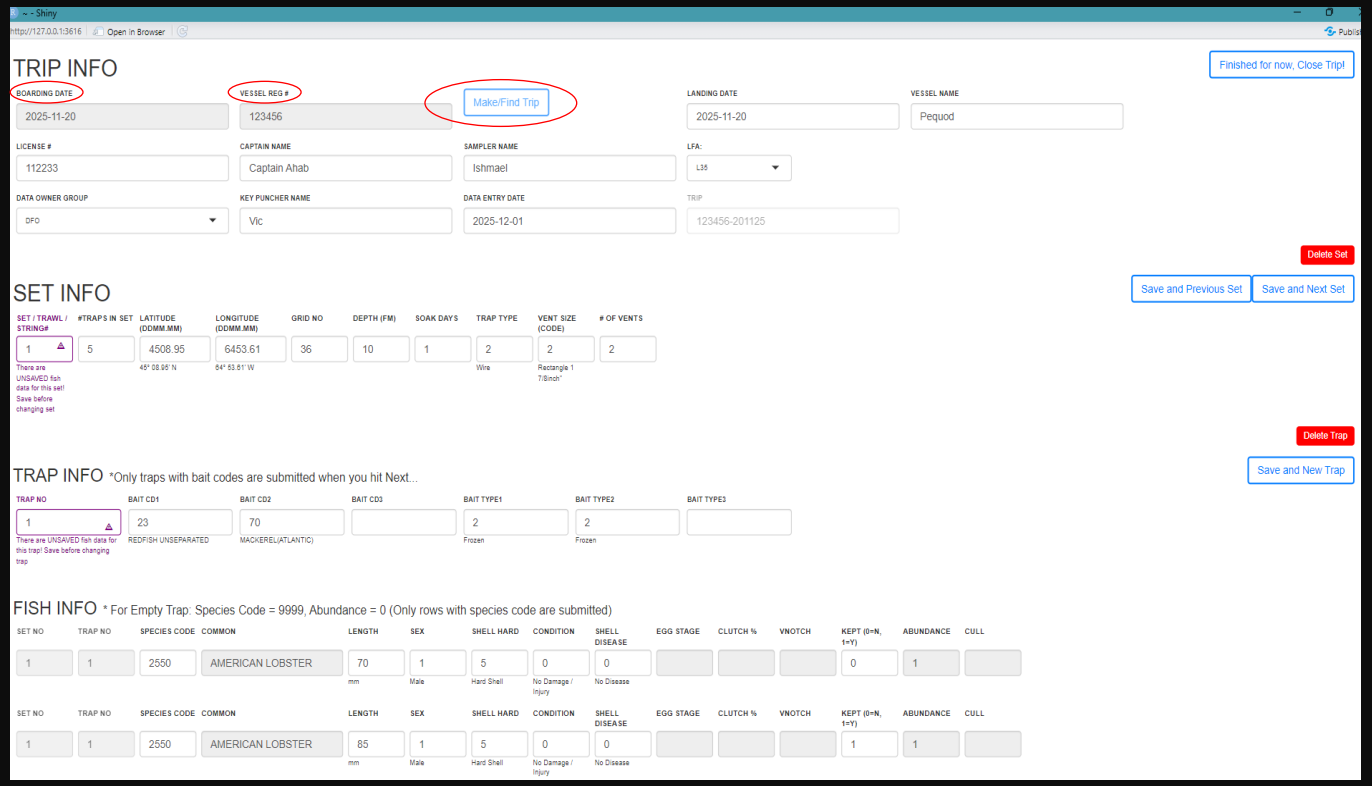
Sample Information

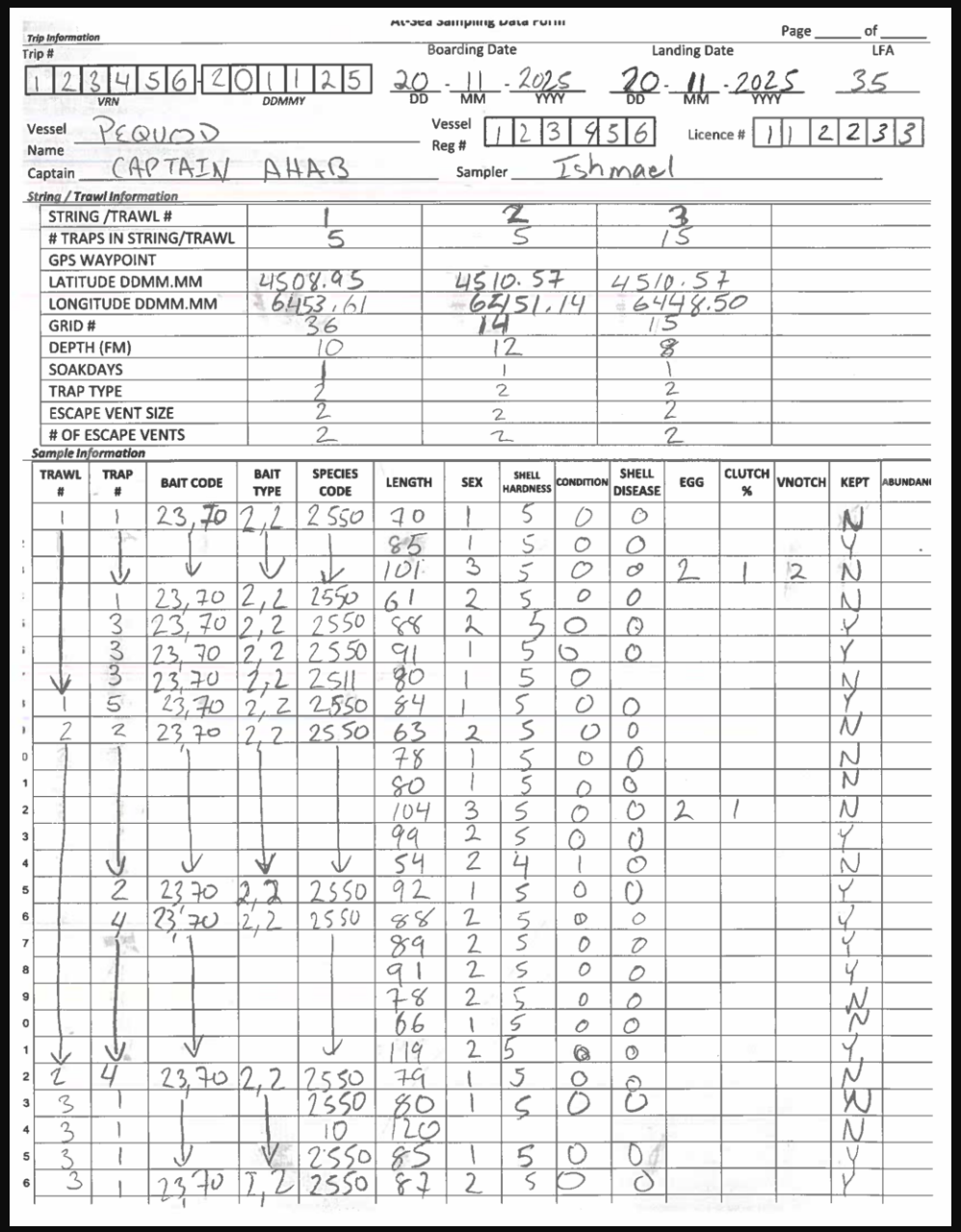
	TRAWL #	TRAP #	BAIT CODE	BAIT TYPE	SPECIES CODE	LENGTH	SEX	SHELL HARDNESS	CONDITION	SHELL DISEASE	EGG	CLUTCH %	VNOTCH	KEPT	ABUNDANCE
1															
2															
3															
4															
5															

Enter Your Data or Look-up Entered Trip:

1. You must first enter the Boarding Date and Vessel Registration Number (VRN) then click the Create/Find Trip button to begin entering data for your trip.

2. If the folder you chose already contains a trip data file with this info, then the trip will be accessed to continue editing.





Enter Your Data:

1. Complete the **TRIP INFO** Section information.

2. Enter **SET INFO**. The Application will check the lat / long are within the boundaries of the LFA listed in the **TRIP INFO** section. A warning will appear if they are not. Check the data, sometimes this warning will appear if fishing is occurring on and near the boundary of the LFA. If the lat / long are correct ignore the warning.

3. Enter the **TRAP INFO.** If there are two Bait Codes (i.e. Redfish and Mackerel) there should also be two Bait Types Entered as well (i.e. Frozen and Frozen).

4. Proceed to enter **FISH INFO** for that trap. Refer to the sampling protocols for the required data for each species group. The system works on species codes rather than names to avoid different names used for the same species in different areas. The app will open the fields that are required, or where data can be entered depending on the species code entered.

5. If an invalid number is entered in a field, the field will be red and you will not be able to save the data.

SET NO	TRAP NO	SPECIES CODE	COMMON	LENGTH	SEX	SHELL HARD	CONDITION
1	3	2511	JONAH CRAB	100 mm	5 Allowed sex values are 1,2,3		78 Not a valid condition code

FISH INFO * For Empty Trap: Species Code = 9999, Abundance = 0 (Only rows with species code are submitted)

SET NO	TRAP NO	SPECIES CODE	COMMON	LENGTH	SEX	SHELL HARD	CONDITION	SHELL DISEASE	EGG STAGE	CLUTCH %	VNOTCH	KEPT (0=N, 1=Y)	ABUNDANCE	CULL
1	3	2550	AMERICAN LOBSTER	88 mm	2 Female	5 Hard Shell	0 No Damage / Injury	0 No Disease				1	1	
1	3	2550	AMERICAN LOBSTER	91 mm	1 Male	5 Hard Shell	0 No Damage / Injury	0 No Disease				1	1	
1	3	2511	JONAH CRAB	100 mm	Missing Value!		Missing value!						1	
1	3	2550	AMERICAN LOBSTER	84 mm	1 Male	5 Hard Shell	0 No Damage / Injury	0 No Disease				1	1	
1	3	10	COD(ATLANTIC)	Missing Value!			Missing value!						1	

Enter Your Data:

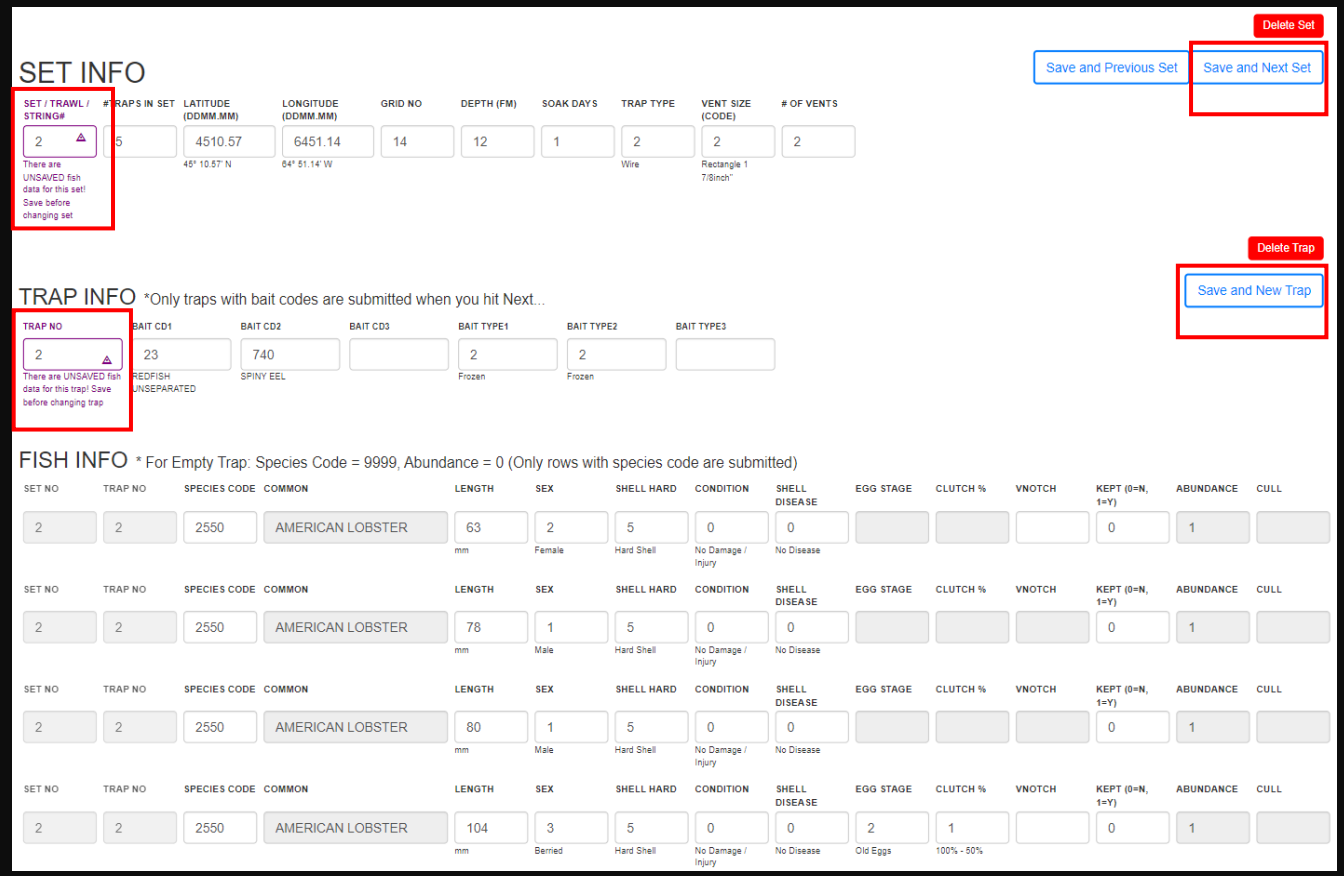
1. After all **FISH INFO** is entered for each **TRAP,** review entered data carefully and use the "Save and New Trap" button to move to the next trap.

2. Double check the String for each Trap (on the data sheet) as you move through the data to avoid entering the Traps under the wrong string.

3. When you are ready to move to the next Trap, use the "Save and New Trap" button, and if the next sampled trap is not the next consecutive number, modify it to match the data set in the "TRAP NO" column.

4. When you are ready to enter a new String, use "Save and Next Set" button.

5. If your next String/Set is not the next consecutive number just modify the number in the "SET/TRAWL/STRING #" column to match the next String on your data sheet.

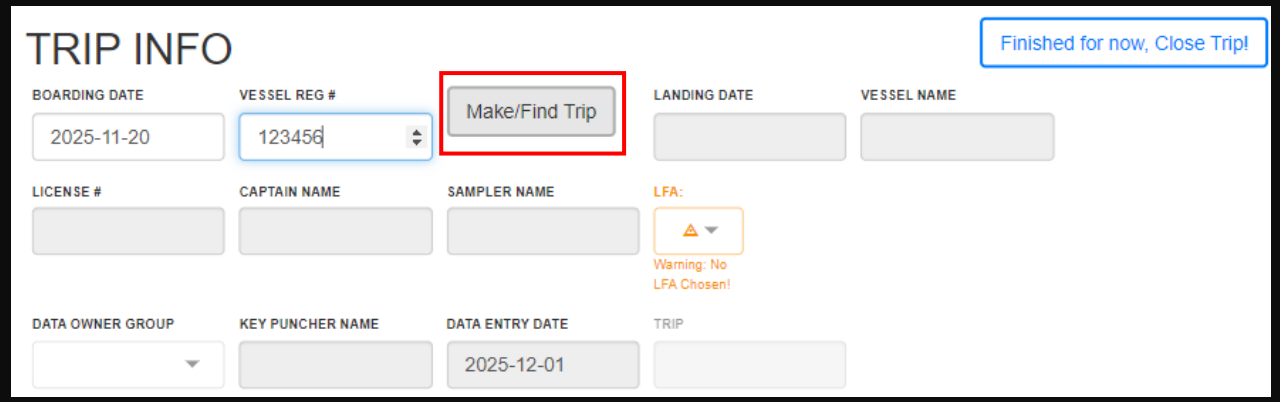


Saving Data to Finish Keypunching Later

1. If you need to save and close the Trip to complete data entry later you can use the “Finished for now, Close Trip!” button.

2. To access the trip you want to continue keypunching you will need to put in the Boarding Date and Vessel Reg # and use the “Make/Find Trip” button.

3. When you are prompted to pick a folder when using input.data() you will need to pick the folder where the trip was previously saved.



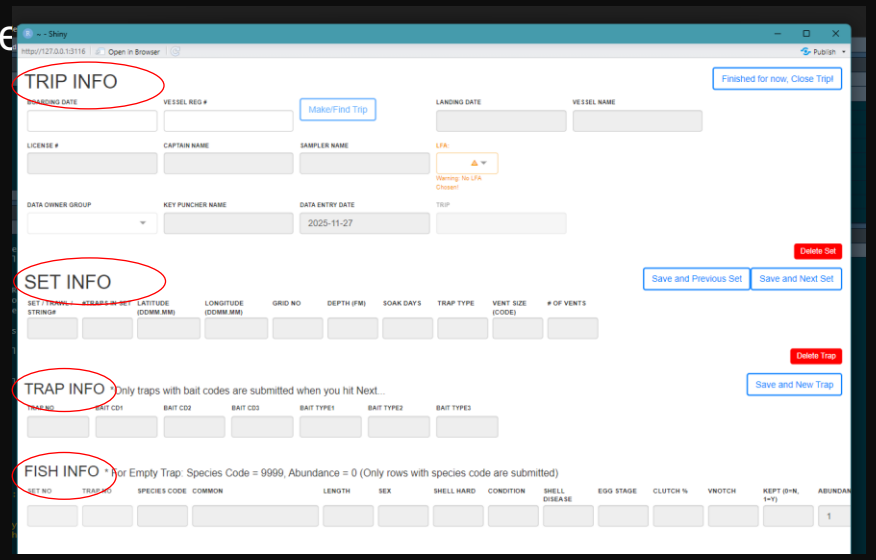
Data Entry Tips:

- ✓ A data entry stand is helpful to avoid losing your spot.
- ✓ **Make note or flag when the string changes ahead of time**. The app is building relational tables, if you enter a trap under the wrong string you will have to delete the traps and all the fish data back to where the string number should have changed, start a new string and re-enter the trap and fish data.
- ✓ Before proceeding to the next Section (Trip, Set, Trap, Fish) check all your entries to avoid having to edit or delete any data.

Review Entered Data

1. Open R Studio, Load the at.sea.lobster package
 > require(at.sea.lobster)

2. Review all trips to ensure they have been entered correctly, and there

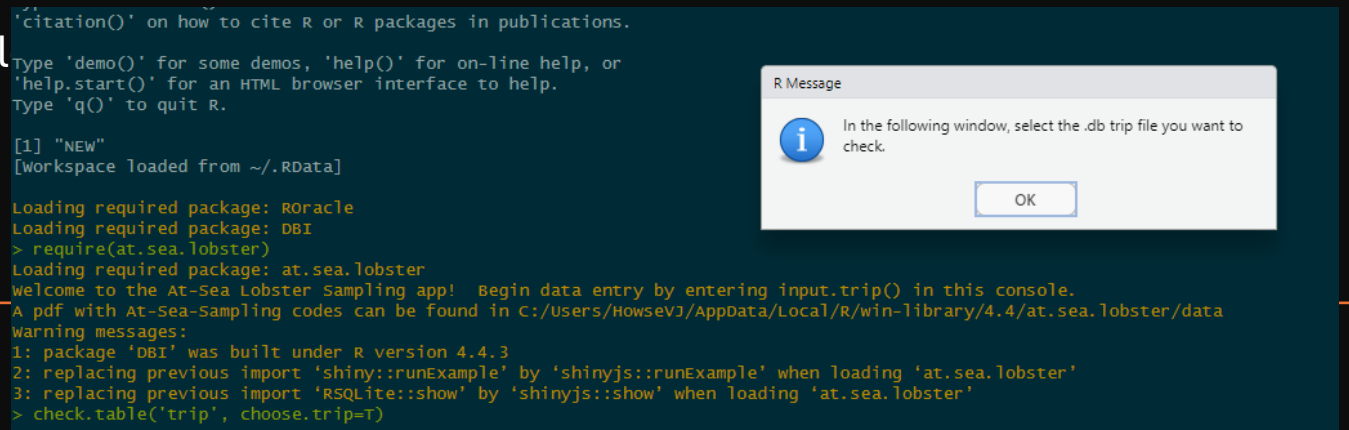


3. Review each Table individually – this is how they are stored:
 - TRIP INFO
 - SET INFO
 - TRAP INFO
 - FISH INFO

To Look at the TRIP:

>check.table('trip', choose.trip=T)

A window will pop up allowed you to pick the fol



Select your trip.

Note: To make this step quicker pin the folder location to the sidebar.

Reviewing SET INFO:

Note: If you don't include "choose.trip=T" R will reopen the last trip that was opened in the data entry screen, not the last trip you opened in the R View with Check.Table()

>check.table('set', choose.trip=T)

When reviewing sets – check that the last set has data.

If it is all NA's the set should be removed.

When entering data if an additional set is keypunched and no traps or fish are entered the final set can still be saved without information attached to it.

Filter

	TRIP_ID	FISHSET_ID	SET_NO	SETCD_ID	GEAR_ID	SPECSCD_ID	STRATUM_ID	EST_CATCH	NUM_TRAPS	LATDDMM	LONGDDMM	DEPTH	SOAK_DAYS	SOURCE	NUM_HOOK	TRAP_
1	123456_201125	123456_201125_1	1	NA	NA	2550	36	NA	5	4508.95	6453.61	10	1	NA	NA	2
2	123456_201125	123456_201125_2	2	NA	NA	2550	14	NA	5	4510.57	6451.14	12	1	NA	NA	2

howing 1 to 2 of 2 entries, 18 total columns

Console Terminal Background Jobs

R 4.4.2 · ~/

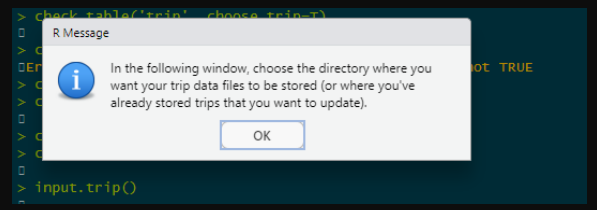
```
.oading required package: at.sea.lobster
velcome to the At-Sea Lobster Sampling app!  Begin data entry by entering input.trip() in this console.
\ pdf with At-Sea-Sampling codes can be found in C:/Users/HowseVJ/AppData/Local/R/win-library/4.4/at.sea.lobster/data
varning messages:
.: package 'DBI' was built under R version 4.4.3
2: replacing previous import 'shiny::runExample' by 'shinyjs::runExample' when loading 'at.sea.lobster'
3: replacing previous import 'RSQLite::show' by 'shinyjs::show' when loading 'at.sea.lobster'
> check.table('trip', choose.trip=T)
>
> check.table('set', choose.trip = T)
>
```

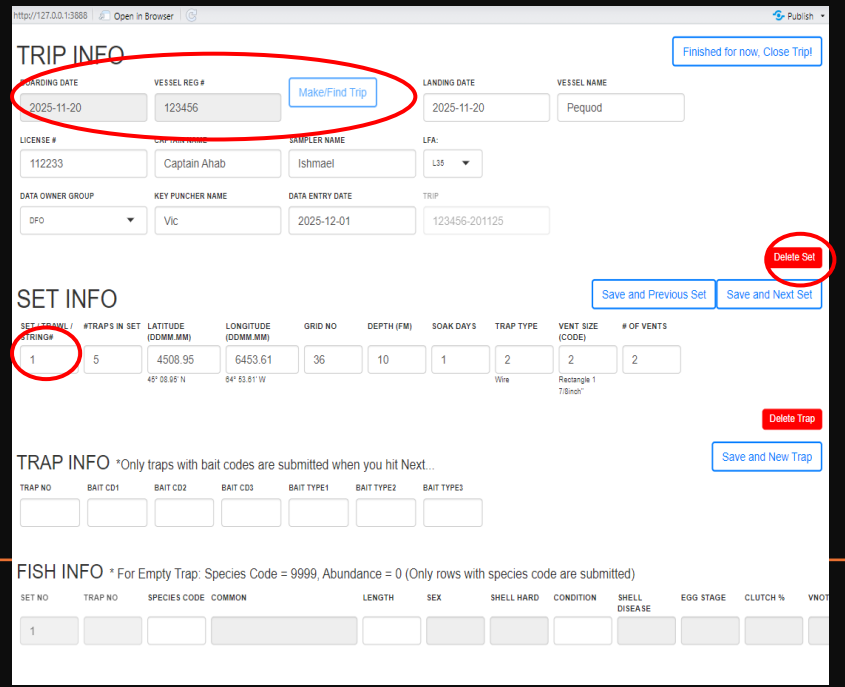
SET_NO	SETCD_ID	GEAR_ID	SPECSCD_ID	STRATUM_ID	EST_CATCH	NUM_TRAPS	LATDDMM	LONGDDMM	DEPTH	SOAK_DAYS	SOURCE	NUM_HOOK	TRAP_TYPE	VENT_CD	NUM_VENTS
22	NA	NA	2550	47	NA	12	4458.594	6517.524	36	1	NA	NA	3	1	2
23	NA	NA	2550	47	NA	12	4458.554	6516.615	24	1	NA	NA	3	1	2
24	NA	NA	2550	35	NA	12	4500.375	6515.094	34	1	NA	NA	3	1	2
25	NA	NA	2550	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

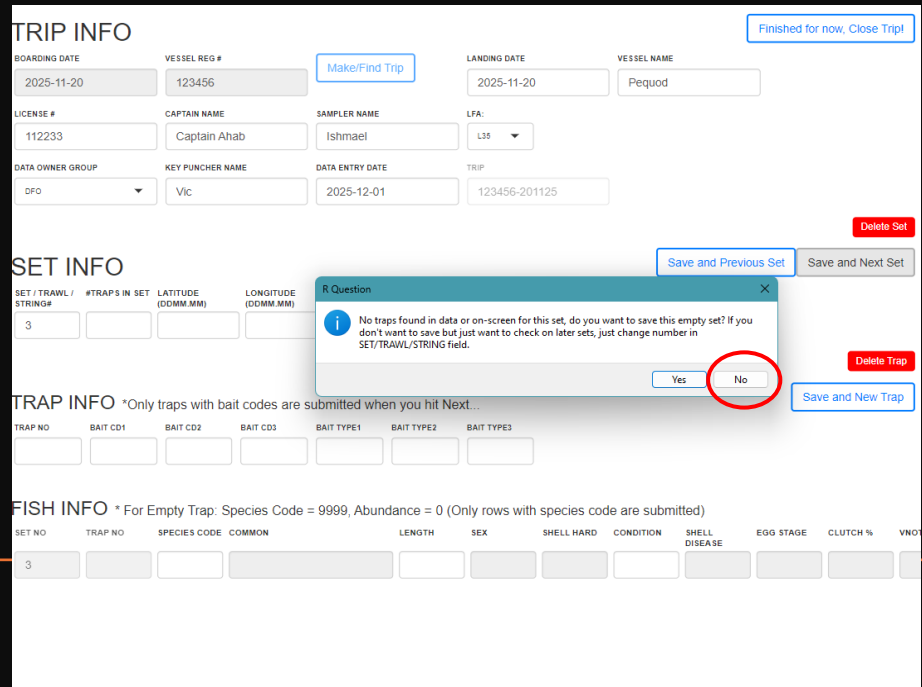
Delete an empty set

1. First open the Shiny App through R Studio
 > require(at.sea.lobster)
 > input.trip()

2. Select the folder and the data entry window will open.
3. input the boarding date and the vessel reg # (found in the TRIP_ID) and click “Make/Find Trip”
4. Enter the set that has no data – and delete it.
5. Select “Finished for now, Close Trip!” and select “No” so that it doesn’t re-save the empty set with NAs again.
6. Re-open the Trip “Set” table an ensure the empty set was deleted.







Check the Trap and Fish Tables

Use the same approach as the Set and Trip checking

```
> check.table('trap', choose.trip=T)
```

```
> check.table('fish', choose.trip=T)
```

REMINDER:

Anything you delete will also delete anything that comes after it

If you delete the last set that was saved that has no Trap or Fish Data – that's ok you're only removing the empty Set.

EXAMPLE:
SET 21 was an accident and has no data – all NAs when you look at SET 21

SET 20 has Traps and Fish info

If you delete SET 20 then you will lose the TRAPS and all FISH Data in and after SET 20, but nothing before it.

Uninstall and Reinstall when updates are needed

If changes are made to the app and it requires updating follow these steps to ensure you have the newest version.

***(any line starting with > is code; only type everything after the >)**

1. Open R Studio
   >remove.packages("at.sea.lobster")

2. Confirm the package is removed
   > library(at.sea.lobster)

   *You should see an error if it's uninstalled.

3. Install the at.sea.lobster App and load package
   > devtools::install_github("LobsterScience/at.sea.lobster")
   > require(at.sea.lobster)